DDD-C-001559(GSA-FSS)
April 14, 1969

INTERIM FEDERAL SPECIFICATION

CARPET, LOOP, LOW PILE HEIGHT, HIGH DENSITY, WOVEN OR TUFTED

WITH ATTACHED CUSHIONING

This Interim Federal Specification was developed by the Federal Supply Service, General Services Administration, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for woven and tufted, low pile height, high density, level loop, carpeting with attached rubber or vinyl cushioning.

1.2 Classification. The carpet covered by this specification shall be of the following types and classes, as specified:

Type I - Woven.
  Class 1 - Single level loop pile.
    Subclass A - (16 oz./sq. yd. pile).
    Subclass B - (18 oz./sq. yd. pile).
    Subclass C - (20 oz./sq. yd pile).

  Class 2 - Multilevel loop pile.

Type II - Tufted
  Class 1 - Single leven loop pile.
    Subclass A - (16 oz./sq. yd. pile).
    Subclass B - (18 oz./sq. yd pile).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

Federal Specifications:
UU-P-268 - Paper, Kraft, Untreated, Wrapping.
CCC-T-191 - Textile Test Methods.
PPP-B-35 - Bags: Textile, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall

Federal Standards:
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 501 - Floor Coverings, Resilient, Nontextile Sampling and Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

FSC 7220

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:
MIL-I-10547 - Liners, Case, and Sheet Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D C. 20580.)

American Society for Testing and Materials (ASTM) Standards.

D-418-68 - Woven Pile Floor Coverings.
D-1335-67 - Tuft Bind of Pile Floor Coverings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Freight Classification Committee, 202 Union Station, Chicago Ill. 60606.)

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Classification Board, 1616 P St , Nw, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified (6.2), a preproduction sample of the carpet, measuring 12 inches by 12 inches, shall be submitted to the contracting officer for approval.

3.2 Standard sample. When a standard sample is available the finished carpets shall match the standard sample for color and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (6.3)

3.3 Material.

3.3.1 Pile yarn. Unless otherwise specified (6.2), the pile yarn shall be made of 100 percent continuous filament nylon. The filament nylon shall be continuous filament high bulk or textured carpet type yarn. Individual filament size shall average 15 denier or coarser The finished yarn shall contain not more than 2.0 percent chloroform soluble material when tested as specified in 4.3.1.

3.3.2 Chain, filling and stuffer yarns. Chain, filling and stuffer yarns, as applicable, used in woven carpets shall be made of vegetable or synthetic fibers. At the option of the contracting officer, chain, filling and stuffer yarns, as applicable, made of synthetic fibers only may be specified (6.2).

3.3.3 Backing material. The backing material for use in tufted carpets shall be a woven fabric of jute weighing not less than 7.6 ounces per square yard, or a woven or nonwoven fabric of 100 percent polypropylene weighing not less than 3.0 ounces per square yard. The nonwoven fabric may consist of less than 100 percent polypropylene fibers, but shall be not less than 75 percent polypropylene fibers with the balance cellulosic, nylon or polyester fibers. When vinyl cushioning

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is specified, a vinyl coated or impregnated fibrous glass cloth or mesh material, not less than 17.5 by 15.5 mesh size, is acceptable. At the option of the contracting officer, a backing material made of synthetic materials only may be specified (6.2).

3.3.4 Back coating. The back coating compound shall be a synthetic resin or natural or synthetic latex compound. The quantity of back coating shall be that normally used in the supplier's production.

3.3.5 Attached cushioning. The cushioning shall be rubber, classes 1 or 2, or vinyl as specified in the contract or order (6.2).

3.3.5.1 Rubber cushioning. The rubber cushioning shall be class 1 or class 2. The compound used in making the cushioning shall be made from natural or synthetic latex or a mixture of natural and syntnetic latices. The cushioning shall be free of objectionable odor and shall have a skin on the floor side when affixed to the carpet. The rubber cushioning shall meet the following requirements:

a. Cushioning shall average not less than 0.187 inch with thickness differential from one side of the carpet to the other not to exceed 0.030 inch.

b. The weight per square yard shall be not less than 2.37 pounds for class 1 rubber and not less than 3.50 pounds for class 2 rubber.

c. The compressibility shall be not less than 3.5 pounds nor more than 7.5 pounds

d. The compression set shall be not more than 15.0 percent for class 1 rubber and not more than 7.5 percent for class 2 rubber.

e Cushioning shall tear before pulling free from carpet in peel test.

f. When subjected to an accelerated aging test, the cushion shall not deteriorate.

3.3.5.2 Vinyl cushioning Vinyl cushioning shall be suitably compounded virgin polymer or copolymer of vinyl chloride resin, plasticized with phosphate or phthalate ester plasticizers only. The compound shall be uniform, free from fiber and objectionable odor. Virgin polymer shall be defined as a polymer or copolymer that has not been processed into a finished product prior to use in the cushioning. The vinyl cushioning shall meet the following requirements

a. Thickness shall be as specified in the contract or order (6.2).

b. Breaking strength shall be not less than 20 lbs. in both machine and transverse directions

c. Ultimate elongation of compound shall be not less than 60 percent in both machine and transverse directions.

d When subjected to an accelerated weathering test, the cushion shall not crack, become stiff and brittle, or soft and tacky and shall not change color appreciably when compared to the unexposed sample after 100 hours in the weatherometer.

e. When subjected to a flexibility test at both 70° F. and 30° F., the cushion shall not crack, flake, craze, or show any other indication of failure.

f. The volatile matter shall not exceed 1.0 percent

3.4 Color and matching. The color of the carpet shall be as specified (6.2) and shall match the standard sample, when available, under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin (K., and shall be a good approximation to the standard sample under incandescent lamplight at 2800° K. when tested as specified in 4.5.14.

3.4.1 Colorfastness. The carpet shall show fastness to light and wet-method cleaning equal to or better than the standard sample. When no standard sample is established, the carpet shall show good fastness to wet-method cleaning and light. They shall show a rating of good for light colors after 20 standard fading hours (2000 Langleys), and for dark colors after 40 standard fading hours (4000 Langleys). Testing shall be as specified in 4.4.

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3.5 Physical requirements. The finished carpets shall conform to the physical requirements specified in table I.

TABLE I. Physical requirements 1/

Type	I					II	
Class	1			2		1	
Description	Single level loop pile			Multilevel loop pile		Single level loop pile	
Subclass	A	B	C			A	B
Tufts/sq. in.	56	80	76	74		56	85
Pile weight oz./ sq. yd.	16	18	20	16		16	18
				High	Low		
Pile height, inch							
Min.	0.100	0.100	0.080	0.090	0.050	0.100	0.100
Max.	0.160	0.150	0.150	0.150	0.080	0.160	0.160
Material							
Pile	See 3.3.1	See 3.3.1	See 3.3.1	See 3.3.1		See 3.3.1	See 3.3.1
Backing yarns and material	See 3 3.2	See 3.3.2	See 3.3.2	See 3.3.2		See 3.3.3	See 3.3 3
Back coating	See 3.3.4	See 3.3.4	See 3.3.4	See 3.3.4		See 3.3.4	See 3.3.4
Pile yarn denier	3000	3000	2000	3000		3000	2/
Ply or yarn twist (turns per inch)							
Min.	0.5	0 5	0.5	0.5		0.5	0.5
Max.	2.0	2.0	2.0	2.0		2.0	2.0
Tuft bind (oz.)	150	200	200	200		150	200

1/ The weights given in the table are exclusive of back coating Tolerance of minus 4.0 percent will be allowed for pile weight only All other requirements shown, except as specified, are minimum This does not prohibit the increase of any or all constituents listed.

2/ 2000 or 3000 denier as specified in the contract or order.

3.6 Shrinkage. The shrinkage shall not exceed 2.0 percent in either the length or the width when tested as specified in 4.4.

3.7 Construction. All carpeting shall be coated on the back or floor side, prior to attachment of the cushioning, with the coating compound specified in 3 3.4. Construction shall conform to requirements specified in table I Cushioning shall cover the entire carpet. All carpet edges shall be straight.

3.7.1 Seaming. Standard widths shall be used unless the size required necessitates seaming. Seams shall be kept to the minimum practicable. Unless otherwise specified, seaming shall be as follows Seams on carpets not installed wall-to-wall, shall be made with carpet type, single face, pressure sensitive adhesive tape with a minimum width of three inches The tape used should be that recommended by the carpet manufacturer, or equal In addition, adhesive shall be applied to both edges of the carpet at the joint These edges shall be brought together to insure direct contact of the adjoining edges after application of the adhesive. For seams on carpets used in wall-to-wall installation the edges shall be brought together to insure good contact. Unless otherwise specified by the ordering activity, the contractor shall secure seams by tape or adhesive, or a combination of these Such tape or adhesive shall be those recommended by the carpet manufacturer, or equal. The finished seams shall be secure and able to withstand conventional commercial rug cleaning processes.

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3.8 Size and pattern. The length, width and pattern of the carpets shall be as specified (6.2).

3.8.1 Dimensional tolerance. The dimensional tolerances shall be as follows:

Cut to size - width or length - not more than 1 percent less than specified.

Rolls - width - not more than 1 percent less than specified; length - not less than specified nor more than 10 percent longer than specified.

Individual roll - not more than 1 yard less than indicated on piece ticket.

Total yardage in sample - total gross length of all pieces in sample not less than the total gross length marked on piece tickets.

3.9 Flame resistance Unless otherwise specified (6.2), the flame resistance of carpets shall be such that the longest diameter of the charred area shall not exceed 2 inches when tested for flame resistance as specified in 4.5.13 (see 6.4).

3.10 Fiber identification. Carpets shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.11 Workmanship. Carpets shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQL's).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required, inspection shall be made of the completed end item for conformance with all provisions of this specification.

4.3 Inspection for acceptance. Inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Inspection of components and materials. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials meet the requirements of this specification. Testing shall be performed on the components listed in table II, as applicable. Except as otherwise specified, test methods referenced are specified in CCC-T-191. The sample unit shall be of sufficient size to conduct the required tests. The sample size shall be as specified in table III. The lot shall be unacceptable if one or more sample units fail to meet any specified requirement.

TABLE II Component testing 1/

Component	Characteristic	Requirement Paragraph	Test Method
Pile yarn (lbs.)	Material identification.	3.3.1	2/
	Denier.	Table I	4021
	Turns per inch.	Table I	4054
	Chloroform sol. material.	3.3.1	2611
Chain, filling and stuffer yarns (lbs ea.), as applicable	Material identification	3.3.2	2/
Backing material (yds.), as applicable	Material identification.	3.3.3	2/
	Weight.	3.3.3	5041
Back coating compound (pt.)	Material identification	3.3.4	2/

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TABLE II. Component testing 1/ (Con't)

Component	Characteristic	Requirement Paragraph	Test Method
Cushioning, rubber:			
Classes 1 and 2 (lbs.)	Material identification.	3.3.5.1	2/
Class 2 (sq. yd.)	Thickness.	3.3.5.1	4.5.4.2
	Weight.	3.3.5.1	4.5.5.2
	Compressibility.	3.3.5.1	4.5.6.2
	Compression set.	3.3.5.1	4.5.7.2
Cushioning, vinyl (lbs.)	Material identification	3.3.5.2	2/

1/ May be accepted based on supplier's certificate of compliance for all characteristics listed
2/ Standard Commercial.

TABLE III. Sample size

Lot size (units)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

4.3.2 Examination of the end item. Defects found during the visual examination shall be classified in accordance with 4.3.2.1, 4.3.2.2 and 4.3.2.3.

4.3.2.1 Visual examination The defects listed in table IV shall be counted regardless of their proximity to one another, except where two or more defects represent a single local condition of the carpet, in which case only the more serious defect shall be counted. Continuous defects shall be counted as one defect for each yard or fraction thereof in which they occur. The sample unit shall be 1 linear yard The acceptable quality levels (AQL's) shall be 2.5 major and 6.5 total defects (major and minor combined) per hundred units (linear yards). The sample size shall be based on inspection level II of MIL-STD-105. The lot size shall be expressed in units of 1 linear yard. When examination is made of full rolls, not more than one third of the total sample yardage shall be examined in any one roll When the lot consists of less than 5 rolls, an approximate equal number of yards shall be examined on each roll to yield the sample yardage When examination is made on carpets cut to size, the number of units selected shall be sufficient to yield the sample yardage. Visual examination shall be made at a distance of approximately six feet.

TABLE IV. Classification of defects

Defects	Classification Major	Minor
Spot or stain		X
Not evenly constructed, affecting appearance.		X
Marks across carpet.	X	
Discolored areas affecting appearance		X
Any noticeable unevenness on top surface affecting appearance.		X
Any obviously objectionable streak lengthwise of weave	X	
Length of pieces or width not as specified.	X	
Tufts missing in pile.	X	
Cut, hole or tear	X	
Seams (when specified) not properly sewed, or taped.	X	
Cushioning imperfectly applied, poor adherence.	X	
Imbedded foreign matter		X

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4.3.2.2 Overall examination. Each defect listed shall be counted not more than once in each unit examined. The sample unit for this examination shall be one roll or cut piece. The number of rolls or cut pieces examined for visual examination shall be the sample size. The lot shall be unacceptable if one or more of the following defects are found:

Overall uncleanness.
Color or pattern other than specified.
Objectionable odor.
Type not as specified.
Edges not finished as required.

4.3.2.3 Length examination (rolls only). Each roll shall be examined for gross length. Any gross length found to be less than the gross length marked on the roll ticket shall be considered a defect with respect to length. The sample unit shall be one roll The rolls used for visual examination shall be used for this examination The lot shall be unacceptable if one or more defects are found in the sample.

4.4 Testing of the end item. Tests shall be performed for the characteristics specified in table V. Except as otherwise specified, test methods referenced are specified in CCC-T-191. Unless otherwise specified, the physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes in the applicable test method. The sample unit shall be 1 square yard for the rolls or one carpet cut to size, as applicable. The lot size for purposes of end item testing shall be expressed in units of square yards for rolls and pieces of carpet, as applicable. The sample size shall be as listed in table III. The lot shall be unacceptable if one or more units fail to meet any requirement specified.

TABLE V. End item testing 1/

Characteristic	Requirement Reference	Test Method
Tufts per square inch.	Table I	4.5.1
Weight of pile yarn.	Table I	4.5.2
Pile height.	Table I	4.5.3
Tuft bind.	Table I	ASTM D-1335-67
Attached rubber cushioning.		
Thickness (Class 1) 7/	3.3.5.1	4.5.4.1
Weight (class 1)	3.3.5.1	4.5.5.1
Compressibility (Class 1)	3.3.5.1	4.5.6.1
Compression set (Class 1)	3.3.5.1	4.5.7.1
Adherence (Classes 1 and 2)	3.3.5.1	4.5.8
Accelerated aging	3.3.5.1	4.5.9
Attached vinyl cushioning. 6/		
Thickness 7/	3.3.5.2	21213/
Breaking strength	3.3.5.2	4.5.10
Elongation	3.3.5.2	4.5.10
Accelerated weathering	3.3.5.2	5804
Flexibility		
70° F	3.3.5.2	4.5.11
30° F	3.3.5.2	65113/4/
Volatile matter	3.3.5.2	92113/5/
Colorfastness		
to light	3.4.1	56602/
to wet method cleaning	3.4.1	4.5.12
Shrinkage	3.6	4.5.12
Flame resistance	3.9	4.5.13

1/ May be accepted based on supplier's certificate of compliance for all characteristics listed
2/ In the event of a dispute resulting from the test with method 5660, or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer shall authorize the exposure to natural light in accordance with method 5662.
3/ Fed. Test Method Std. No. 501.
4/ Low temperature flexibility shall be tested using a 1 ± 1/8 inch mandrel.
5/ The oven should be regulated to insure minimum air velocity.
6/ Results of tests conducted on vinyl cushioning prior to attachment to carpet may be accepted.
7/ The test results for each of these characteristics shall be reported as the lot average.

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4.5 Tests.

4.5.1 Tufts per square inch. Tufts per square inch shall be determined by multiplying the pitch per inch of pile yarn ends by the rows of pile per inch, or the equivalent thereof. Pitch and rows per inch shall be determined in accordance with ASTM D-418-68. Pitch is defined as the number of pile yarn ends per inch of width in the carpet. Rows is defined as the number of rows of tufts per inch of length (also referred to as stitches per inch for tufted).

4.5.2 Weight of pile yarn. The pile yarn shall be separated from other yarns (prior to attachment of cushioning) on a sample containing 16 square inches. To express weight of pile yarn per carpet, the weight of the pile shall be calculated as follows Weight of pile yarn (ounces) in 16 square inch sample X 81.0 = weight of pile yarn in ounces per square yard of carpet.

4.5.3 Pile height. The pile height shall be determined in accordance with ASTM D-418-68 For multilevel carpet, use the following method A metal rule graduated in 1/100 inch approximately 3/4 inch wide and 0.040 inch thick shall be inserted between the lengthwise pile rows. The height of the pile in the high pile areas and in the low pile areas shall each be measured five times Each of the measurements shall be at least 1 inch apart. The average of the five measurements in the high pile area shall be the high pile height and the average of the five measurements in the low pile area shall be the low pile height.

4.5.4 Thickness of rubber cushioning.

4.5.4.1 Class 1 rubber cushioning. The specimen shall be one square yard of the finished item (with cushioning). The thickness between the two plane surfaces of the specimen shall be determined under a pressure of 0.100 pounds per square inch (p.s.i.) (± 0.001 p.s.i.) distributed over a circular area 3.00 inches ± 0.01 inch in diameter Apply pressure slowly to avoid impact and protect the specimen from vibration during the test. Five readings shall be taken on the specimen and the average computed to the nearest 0.001 inch (millimeter). The results from the sample units in the sample size shall be averaged and computed to the nearest 0.001 inch (millimeter). This is the lot average thickness of the finished item A one square yard specimen of the carpet before application of cushioning shall be tested for thickness as specified above The results from the sample units in the sample size shall be averaged and computed to the nearest 0.001 inch (millimeter) This is the lot average thickness of the carpets before application of the cushioning The difference between the thickness before application of the cushioning and the thickness of the finished item shall be the average thickness of the rubber cushioning

4.5.4.2 Class 2 rubber cushioning The specimen shall be one square yard of the class 2 rubber cushioning The thickness shall be determined as specified in 4.5.4.1 and the results from each sample unit reported separately.

4.5.5 Weight of the rubber cushioning

4.5.5.1 Class 1 rubber cushioning The specimen shall be one square yard of the finished item (with cushioning) The specimen shall be conditioned for a minimum of 4 hours under standard conditions of 70° F (± 2° F) and 65 percent (± 2 percent) relative humidity for testing The specimen shall be weighed to the nearest 0.01 pound per sq. yard The results from the sample units in the sample size shall be averaged and computed to the nearest 0.01 pound per square yard This is the lot average weight of the finished item. A one square yard specimen of the carpet before application of the cushioning, shall be tested for weight as specified above The results from the sample units in the sample size shall be averaged and computed to the nearest 0.01 pounds per square yard This is the lot average weight of the carpet before application of the cushioning The difference between the weight before application of the cushioning and the weight of the finished item shall be the lot average weight of the cushioning and shall be reported to the nearest 0.01 p.s.y.

4.5.5.2 Class 2 rubber cushioning Specimen shall be one square yard of the class 2 rubber cushioning The weight shall be determined as specified in 4.5.5.1 and the results from each sample unit reported separately to the nearest 0.01 pound per sq. yd.

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4.5.6 Compressibility.

4.5.6.1 Class 1 rubber cushioning. The specimen shall be a one-inch (2.5 cm.) square sample, the thickness of the rubber cushioning, secured by dissecting the carpet materials using care to minimize the removal of rubber with the embedded carpet material. The specimen shall be compressed to 75 percent of its original thickness Thickness shall be measured with a gauge having a circular foot one square inch (6.5 square centimeters) area under a load of 100 grams (g.) (0.220 lbs.). The number of pounds required to compress the sample shall be the compressibility and shall be reported to the nearest pound.

4.5.6.2 Class 2 rubber cushioning. The compressibility of the class 2 cushioning shall be determined as specified in 4.5.6.1 except that the specimen shall be a one-inch (2.5 cm) square sample cut from the full thickness of cushioning material before adhering to the carpet.

4.5.7 Compression set.

4.5.7.1 Class 1 rubber cushioning. The specimen shall be a 2 by 2 inch square, the thickness of the rubber cushioning, secured in accordance with 4.5.6.1. The specimen shall be compressed 50 percent of its original thickness between two parallel plates. The thickness shall be measured in accordance with 4.5.6.1. The sample compressed shall be placed in a Geer oven at 158° F. (± 2° F.) for 22 hours (± 1/2 hour). At the end of the specified time, the sample shall be removed from the plates and allowed to rest at room temperature for 30 minutes. The thickness measurement shall be taken and subtracted from the original thickness The loss in thickness shall be expressed as a percentage of the original thickness and reported to the nearest percent.

4.5 7.2 Class 2 rubber cushioning The compression set of the class 2 cushioning shall be determined as specified in 4.5.7.1 except that the specimen shall be a 2 by 2 inch sample cut from the full thickness of cushioning material before adhering to the carpet.

4 5 8 Adherence Grasp the base carpet with the fingers of one hand and the thickness of the cushioning with the fingers of the other hand and pull firmly in opposite directions. The cushioning should tear before pulling free from the carpet.

4.5.9 Accelerated aging test. A piece of the rubber cushioning shall be placed in an oxygen bomb (not more than 1 ounce of rubber per 170 cubic inches of oxygen) at a temperature of 158°F. and a pressure of 300 pounds (± 10 pounds) per square inch for a period of 7 days. Upon removal, sample should not be sticky and should not crack when bent back upon itself.

4.5.10 Breaking strength and elongation (vinyl cushioning). Breaking strength and elongation shall be determined as described in methods 4111 and 4121 of Fed Test Method Std. No 501. Report shall be made in pounds rather than pounds per square inch. Thickness of sample need not be determined. Die II shall be used in lieu of die III.

4.5.11 Flexibility at 70°F. A specimen of cushioning, not less than 4 by 4 inches shall be doubled and pressed flat on itself in any direction with the topside outside and held in this position for a period of five minutes. At the end of the five minute period the specimen shall be examined for breaking, cracking, or any other evidence of failure, while in the doubled and pressed flat state. The examination shall be visual and at a normal reading distance of one foot. One determination shall be performed for each sample unit and the result shall be expressed as pass or fail

4.5.12 Shrinkage and colorfastness test. A 12 by 12 inch sample of the specified carpet shall be conditioned under standard conditions as defined in CCC-T-191 for a period of 24 hours. Specimen shall then be marked and measured at three different locations in the length and width directions; immerse sample in 110°F. water for 15 minutes. mix 2 g of sodium alkylsulfate type of detergent with 50 g. of water at 110°F. and apply to pile surface of carpet. Scrub sample with a soft bristle brush by stroking back and forth 20 times (10 times in each unilateral direction), and in both length and width directions, rinse well to remove majority of detergent. squeeze and dry at 125°F. until bone dry, again condition under standard conditions for 24 hours

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4.5.12.1 Shrinkage evaluation. The specimen shall be remeasured and the shrinkage computed using the following formula:

$$\text{Shrinkage} = \frac{A-B}{A} \times 100$$

where
A = Average of initial measurement.
B = Average measurements after shampooing.

The shrinkage in both the length and width directions shall be reported to the nearest 0.1 percent.

4.5.12.2 Colorfastness evaluation. Colorfastness evaluation shall be conducted in accordance with method 5610 of CCC-T-191.

4.5.13 Flame resistance. This method covers the procedure for measuring flammability. The test shall be carried out in a suitably sized chamber or box which is open at the top to provide adequate ventilation and which, at the same time, protects the specimen and the ignition flame from drafts. The box should be 12 inches by 12 inches by 9 inches. The ignition media shall be Methenamine Timed Burning Tablet (6.5). The measuring device shall be a scale graduated in inches and tenths of inches. Two test specimens, each 6 inches square, shall be cut from the fabric to be tested. The test shall be done using 30 percent relative humidity. This condition is to be arrived at by using a small laboratory drying oven in conjunction with the standard conditioned testing room. From the psychrometric chart, it can be determined that air at 70°F and 65 percent relative humidity will have a relative humidity of 30 percent when heated to 94°F. Set the oven controls to maintain this latter temperature, place the samples in the oven and allow them to come to equilibrium. Without disturbing the configuration of the pile, the sample shall then be laid out in the test chamber smoothly, horizontally and without tension. A burning tablet shall then be placed firmly in the center of the sample and ignited by touching a match carefully to the edge of the tablet so as not to contact the surface of the fabric. The ignition flame and any propagated flame shall be allowed to burn to completion. The longest diameter of the charred area shall be measured to the nearest tenth of an inch.

4.5.14 Color matching test. A test specimen at least 8 inches square shall be mounted at an angle of 45° to the horizontal and compared with the standard sample, or the approved color sample furnished, similarly mounted under the light conditions specified in 3.4. When an artificial light source is used it shall be placed so that its rays strike the sample normal to its surface. The viewing distance shall be 18 to 24 inches.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. Carpets, of one description only, shall be rolled tightly on a convolute or spiral wound chip board tube or a pole. The tube shall have a minimum wall thickness 0.25 inch, with a minimum inside diameter of 2 inches. For rolls weighing 200 pounds or more, regardless of width, the tube shall have a minimum wall thickness of 0.375 inch, with a minimum inside diameter of 3 inches. The ends of the tube shall extend not more than 1 inch beyond the rolled carpet. The rolled carpet shall be secured with cotton tape, or twine, wrapped with paper conforming to UU-P-268, grade B, minimum bursting strength not less than 80 pounds (24 by 36-500) and sealed with tape conforming to PPP-T-45, grade B.

5.1.2 Level B. The carpets shall be packaged as specified in 5.1.1.

5.1.3 Level C. The carpets shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Carpets, of one description only, packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-576, class 2; PPP-B-591, class II; PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, class weather resistant, grade V3c, style FOL; or PPP-B-640, Class 2, grade A. Each box shall be provided with a case liner conforming to MIL-L-10547, type I, grade C. Closure and strapping shall be in accordance with the appendix to the box specification. When specified (see 6.2), carpets shall be packed in bags conforming to PPP-B-35, type III, class 2, style B with waterproof sewn seams. The bag shall be closed with two double looped wire ties on each open end. The ties shall be not less than 6 inches long.

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5.2.2 Level B. Carpets, packaged as specified in 5.1, shall not require additional packing.

5.2.3 Level C. The carpets, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery at destination at lowest rates, in containers complying with the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special markings required by the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable.

6. NOTES

6.1 Intended use. The carpeting included in this specification is constructed with extra low pile height, with dense pile, to broaden the selection of carpeting primarily for special purpose use.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following.

(a) Title, number, and date of this specification.
(b) Type, class and subclass required (1.2).
(c) When preproduction sample is required (3.1).
(d) Pile yarn, if other than specified (3.3.1).
(e) When synthetic chain, filling and stuffer yarns or synthetic backing material only is required (3.3.2 and 3.3.3).
(f) Type of cushioning (3.3.5).
(g) Thickness of vinyl cushioning when required (3.3.5.2).
(h) Color required (3.4).
(i) Length, width and pattern of carpet (3.8).
(j) Flame resistance requirement, if other than as specified (3.9)
(k) Selection of applicable levels of packaging and packing (5.1 and 5.2).
(l) When packing of carpets in bags as specified in 5.2.1 is required (5.2.1)

6.3 For access to the standard sample, when available address the procuring office issuing the invitation for bids.

6.4 Special flame resistance. Federal agencies responsible for hospitals and medical facilities are directed to the publication General Standards of Construction and Equipment for Hospital and Medical Facilities, issued by the U.S. Department of Health, Education, and Welfare. This publication contains regulations relating to the implementation of the Hill-Burton program, including special construction and fire-resistive requirements.

6.5 Methenamine Timed Burning Tablet - Eli Lilly #1588 or equal.

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